CYBER SECURITY 101

FIRST BLUE-TEAM HOME LAB

30 days. Free tools. One documented project for your portfolio.

30-DAY WORKBOOK | PDF

Before you start

A 30-day, beginner-friendly plan to build ONE documented blue-team home lab with free tools on the computer you already own, and turn it into a portfolio write-up.
No job or certification outcome is promised. Practise only on machines you own. Tool names belong to their owners; this guide is not affiliated with them.

What you need

A computer with at least 8 GB RAM (16 GB recommended) and ~60 GB free disk.
Free software: VirtualBox (virtualbox.org), Ubuntu Server (ubuntu.com), Wazuh (wazuh.com, open source SIEM/XDR).
Optional: Wireshark (wireshark.org).

Week 1 - Build

Day 1 install VirtualBox. Day 2 create Ubuntu Server VM; take snapshot. Day 3 create a second VM (endpoint). Day 4 host-only network; confirm ping. Day 5 SSH key-only login; disable password auth. Day 6 enable ufw; allow only SSH. Day 7 write up: diagram + what you did.

Week 2 - Collect

Day 8 install Wazuh manager (follow the official quickstart). Day 9 enroll the endpoint agent. Day 10 find auth logs in the dashboard. Day 11 generate 10 failed SSH logins; find the alerts. Day 12 note rule IDs that fired. Day 13 snapshot. Day 14 write up: screenshots + what each alert means.

Week 3 - Detect

Day 15 add a test user; watch the event. Day 16 sudo usage alerts. Day 17 file integrity monitoring on /etc. Day 18 change a file; find the alert. Day 19 capture traffic with Wireshark; identify SSH vs DNS. Day 20 tune one noisy rule. Day 21 write up detection notes.

Week 4 - Investigate & publish

Day 22 run a mini 'incident': brute force + new user. Day 23 triage using the timeline sheet. Day 24 write a 1-page incident report. Day 25 restore from snapshot. Day 26 README for GitHub. Day 27 redact IPs/secrets. Day 28 publish. Day 29 practise explaining it in 2 minutes. Day 30 plan your next lab.

Incident timeline sheet

Time | Source/log | What happened | Evidence | Next action

1-page incident report template

Summary:
Scope (hosts, users):
Timeline:
Root cause:
Containment / fix:
Lessons learned:

Contact

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